

Codefest 5

Hardware for Artificial Intelligence and Machine Learning (HW4AI)

ECE 410/510

Spring 2026

Introduction

Deliverables are committed to your GitHub repository by the following Sunday at 11:59 pm. Each codefest is graded Satisfactory (S) / Not Yet (NY) as a CMAN+CLLM pair. A late submission counts as Not Yet and consumes one revision token.

CMAN — Systolic array trace (weight-stationary)

CMAN No AI	No AI assistance permitted. Using a calculator or spreadsheet is fine.
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A 2×2 weight-stationary systolic array computes $C = A \times B$ where: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, expected $C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$. In weight-stationary dataflow, weights are pre-loaded into PEs and stay fixed; inputs stream in from the left; partial sums accumulate downward.

Tasks

1. Draw the 2×2 array of processing elements (PEs). Label each PE with its preloaded weight: $PE[0][0] = B[0][0] = 5$, $PE[0][1] = B[0][1] = 6$, $PE[1][0] = B[1][0] = 7$, $PE[1][1] = B[1][1] = 8$.
2. Fill in a cycle-by-cycle table (columns: Cycle, Input to row 0, Input to row 1, $PE[0][0]$ partial sum, $PE[0][1]$ partial sum, $PE[1][0]$ partial sum, $PE[1][1]$ partial sum, Output C values). Trace at least 4 cycles.
3. Count: (a) total MAC operations performed; (b) number of times each input value is reused; (c) number of off-chip memory accesses for A, B (as inputs), and C (as output).
4. If this were output-stationary instead, which values would stay fixed in the PEs? Give a one-sentence answer.

CMAN deliverable

Deliverable: `codefest/cf05/cman_systolic_trace.md` (or .pdf/.png) containing

- (a) PE diagram with all 4 preloaded weights labeled;
- (b) complete ≥ 4 -row cycle table with partial sum progression;
- (c) MAC count, input reuse count, off-chip access count for A, B, C;
- (d) one-sentence output-stationary answer.